

Cycle-Consistent Patch-Contrastive Harmonization of Multi-Site Glioma MRI: A Mechanism Study with Downstream Validation

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Abstract. Scanner- and protocol-induced intensity shift confounds multi-institutional MRI studies, and unpaired image translation is an increasingly common remedy. Two largely disjoint mechanisms preserve anatomical content during translation: *global, bidirectional* cycle-consistency, and *local, feature-space* patch-contrastive alignment (PatchNCE). We argue these mechanisms fail in complementary ways under unpaired glioma data—the cycle constraint admits an appearance-leaking, boundary-smoothing equilibrium, while one-sided PatchNCE lacks a global appearance anchor—and we study their combination on a self-attention 2.5D tri-planar backbone. We frame the work as a *mechanism study*: rather than asking “does adding PatchNCE help,” we ask under what weighting and measured by which task the cycle/contrastive combination resolves a failure mode that neither component resolves alone. On a BraTS $\leftrightarrow$ UPenn-GBM benchmark we evaluate with distribution metrics (FID/KID, domain-classifier confusion), brain-masked *windowed* structural similarity, and *downstream* cross-site glioma sub-region segmentation (Dice/HD95), ablating the contrastive weight $\lambda_{\text{NCE}} \in \{0, 0.1, 0.5, 1.0, 2.0\}$ at a matched fine-tuning budget. Across a matched-budget sweep, $\lambda_{\text{NCE}}=0.5$ minimizes Fréchet distance to the target site (47.3 vs. 51.8 for the cycle-only control) and monotonically improves brain-masked structural preservation ($0.29 \rightarrow 0.44$), confirming that patch-contrastive alignment resolves a content-preservation failure of pure cycle-consistency. Yet under a corrected, training-consistent protocol the cross-site segmentation gap is already small (cross-site Dice $\approx$ within-site) and image harmonization does not close it, degrading a real-image-trained segmenter uniformly across λ_{NCE} . Distributional harmonization quality and downstream task utility are therefore distinct axes.

Keywords: MRI harmonization · unpaired translation · contrastive learning · cycle-consistency · downstream validation.

1 Introduction

Pooling magnetic resonance imaging (MRI) across institutions is essential for statistically powered neuro-oncology studies, yet scanner vendor, field strength,

receive-coil configuration, and acquisition protocol induce systematic intensity and contrast variation that confounds quantitative analysis. Feature-space correction such as ComBat [3] removes batch effects from *extracted features* but cannot produce harmonized *images*, and is therefore unavailable to spatial downstream tasks—tumor segmentation, lesion volumetry, voxel-level radiomics—that operate on the image itself. This motivates generative, image-space harmonization, in which an unpaired translation model maps an image acquired at one site to the appearance of another while preserving anatomy.

The central difficulty is preserving content. Two mechanisms dominate the unpaired-translation literature. *Cycle-consistency* [16] enforces content preservation *globally and bidirectionally* through pixel-space reconstruction, but the cycle constraint is satisfiable by an appearance-leaking equilibrium: under unpaired data with pathology, the generator can hide source information in low-amplitude, high-frequency signal that survives the round trip, manifesting as over-smoothed tissue boundaries and, in glioma cohorts, attenuated or hallucinated lesion texture. *Patch-contrastive* translation (CUT/PatchNCE [10]) instead preserves content *locally in feature space* by maximizing patch-level mutual information, which is robust to the cycle’s failure mode—but applied one-sided it provides no global anchor on the target appearance distribution and is known to require aggressive weighting and careful negative sampling to remain stable [14,15]. We hypothesize that these failure modes are *complementary*, and we study their combination on a self-attention 2.5D tri-planar backbone for within-modality glioma harmonization.

A second, methodological point motivates this work. Harmonization quality is routinely reported with image-similarity scores, but a global structural-similarity index over an MRI slice is dominated by the large background region and is misleadingly high; worse, it does not reveal whether disease-relevant signal is preserved. The appropriate test is *downstream*: does a model trained at one site retain its accuracy when applied, after harmonization, to another? We therefore center our evaluation on cross-site tumor-segmentation transfer, and we report distribution-matching and *brain-masked windowed* similarity rather than a single global score.

Contributions.

1. A *mechanism* analysis of cycle-consistency versus patch-contrastive content preservation for within-modality MRI harmonization, with a controlled λ_{NCE} ablation (including the $\lambda_{\text{NCE}}=0$ cycle-only control) at a matched fine-tuning budget that locates the operating regime in which the combination is beneficial.
2. A downstream-validated, training-consistent evaluation protocol on glioma cohorts (BraTS/TCGA-GBM, UPenn-GBM): cross-site sub-region segmentation Dice/HD95, distribution-matching (FID/KID, domain-classifier confusion), and brain-masked windowed similarity. We show that a naive harmonization-for-evaluation pipeline (mismatched channel order and normalization) silently corrupts results, and we describe the consistent protocol that avoids it.

3. An honest, downstream-grounded finding. The patch-contrastive term improves distributional alignment and structural preservation with an interior optimum at $\lambda_{\text{NCE}}=0.5$; yet, measured with the corrected protocol, image harmonization does *not* improve cross-site glioma segmentation for this cohort pair—the raw cross-site gap is already negligible and a real-image-trained segmenter is sensitive to synthesized texture. We argue that distributional quality does not entail downstream benefit, and that downstream-grounded evaluation must be reported even when it disagrees with distribution metrics.

2 Related Work

Contrastive unpaired translation. CUT [10] replaced cycle-consistency with a one-sided multi-layer PatchNCE objective drawing negatives from within the input image; DCLGAN [4] showed contrastive and cycle-style dual objectives are complementary on natural images. Subsequent work refined where and how patches are sampled—query selection [6], semantic-relation consistency [8], hard-negative generation [14], and modulated negatives [15]—all instances of the InfoNCE estimator [9]. We deliberately *re-introduce* full cycle-consistency on top of multi-layer PatchNCE—the inverse of CUT’s design choice—and justify it rather than hide it: cycle supplies the global appearance anchor that one-sided PatchNCE lacks, so the contrastive term can act as a complement at a weight well below the CUT/FastCUT regime.

MRI harmonization. Beyond feature-space ComBat [3], image-space harmonizers include disentanglement methods (CALAMITI [17], HACA3 [18]), the 2.5D VAE-GAN ImUnity [2], and the many-to-one CycleGAN IGUANe [11]. The closest methodological precedent, cycleCUT [13], combines cycle and contrastive losses—but for cross-modality MR→CT synthesis, with no within-modality harmonization and no lesion-level downstream task; one-sided contrastive medical translation otherwise relies on auxiliary segmentation supervision [12]. To our knowledge no prior work studies a cycle + multi-layer-PatchNCE hybrid for within-modality MRI harmonization with glioma downstream validation. We position our work as a mechanism study in that gap rather than as a new loss.

Evaluation. We follow stronger conventions than image similarity alone: distribution matching (FID/KID [5,1], domain-classifier confusion), brain-masked *windowed* similarity rather than a global moment, and downstream segmentation [7], all aggregated at the subject level.

3 Method

Backbone. We build on a self-attention CycleGAN with a 2.5D tri-planar generator: each input stacks three consecutive axial slices (s_{z-1}, s_z, s_{z+1}) across four modalities (FLAIR, T1, T1ce/T1gd, T2), giving inter-slice context at the

SA-CycleGAN-2.5D with PatchNCE Hybrid Loss

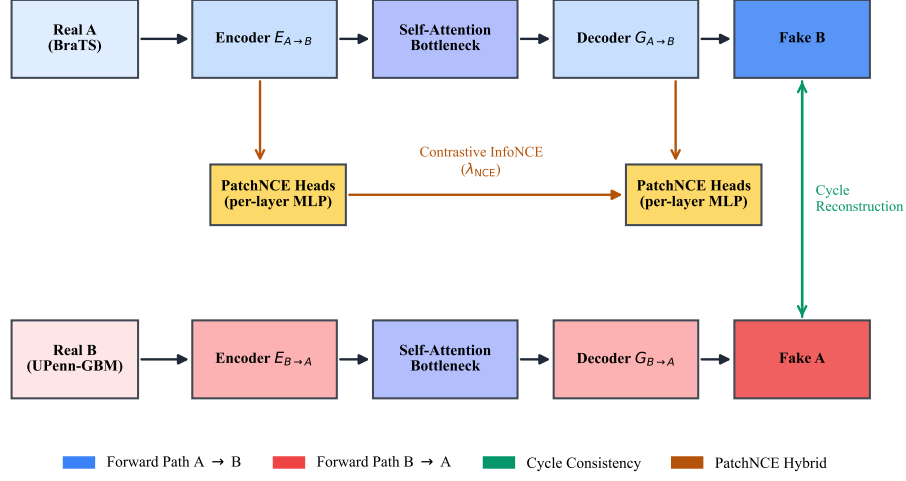


Fig. 1. Self-attention 2.5D CycleGAN with per-layer PatchNCE projection heads. Cycle-consistency provides a global, bidirectional appearance/anatomy anchor; multi-layer PatchNCE provides a local, feature-space content regularizer.

memory cost of a 2D model, with self-attention to capture the spatially non-stationary component of scanner bias. The generators $G_{A \rightarrow B}, G_{B \rightarrow A}$ output the harmonized center slice; the architecture is summarized in Fig. 1.

Cycle-consistency (global anchor). For unpaired collections $\mathcal{D}_A, \mathcal{D}_B$,

$$\mathcal{L}_{\text{cyc}} = \mathbb{E}_{x_a} [\|x_a - G_{B \rightarrow A}(G_{A \rightarrow B}(x_a))\|_1] + \mathbb{E}_{x_b} [\|x_b - G_{A \rightarrow B}(G_{B \rightarrow A}(x_b))\|_1]. \quad (1)$$

Multi-layer PatchNCE (local anchor). Encoder features at L layers are passed through 2-layer MLP heads h_ℓ and L2-normalized. For a query z , its positive z^+ at the corresponding spatial location, and internal negatives z^- , the InfoNCE loss is

$$\ell(z, z^+, z^-) = -\log \frac{\exp(z \cdot z^+ / \tau)}{\exp(z \cdot z^+ / \tau) + \sum_n \exp(z \cdot z_n^- / \tau)}, \quad \mathcal{L}_{\text{NCE}} = \mathbb{E} \sum_{\ell=1}^L \sum_s \ell(\cdot), \quad (2)$$

with temperature $\tau=0.07$, $L=5$ encoder stages, and 256 sampled locations per layer, following CUT [10].

Hybrid objective. PatchNCE *augments*, rather than replaces, cycle-consistency:

$$\mathcal{L} = \mathcal{L}_{\text{adv}} + \mathcal{L}_{\text{cyc}} + \mathcal{L}_{\text{idt}} + \mathcal{L}_{\text{ssim}} + \lambda_{\text{NCE}} \mathcal{L}_{\text{NCE}}. \quad (3)$$

We sweep $\lambda_{\text{NCE}} \in \{0, 0.1, 0.5, 1.0, 2.0\}$ at otherwise identical hyperparameters, where $\lambda_{\text{NCE}}=0$ is the cycle-only control. A selected weight below the CUT/FastCUT regime (1.0/10.0) is principled rather than incidental: because cycle-consistency already supplies a strong content constraint, the contrastive term should complement it, not dominate; over-weighting re-introduces the instability that PatchNCE exhibits when it must carry content alone.

4 Experimental Setup

Data. Site A is BraTS/TCGA-GBM (88 subjects) and site B is UPenn-GBM (566 subjects), each with four co-registered modalities and expert tumor segmentations at 1 mm isotropic, intensity-normalized to $[0, 1]$. We split *by subject* 80/10/10 and sample axial 2.5D triplets over the central axial range; all metrics are aggregated to the subject level.

Training. Each λ_{NCE} arm is warm-started from a converged base SA-CycleGAN and fine-tuned for a matched budget (Adam, $\beta_1=0.5$; mixed precision; A100), so that any difference across arms reflects the contrastive weight and not the optimization budget.

Metrics. Distribution matching: FID/KID (harmonized vs target, per modality), domain- classifier accuracy toward chance, and feature-space MMD; structure: brain-masked *windowed* SSIM (a global moment is misleading here); downstream: a U-Net trained on one site and applied cross-site, raw versus harmonized (whole-tumor and sub-region Dice, HD95). *Critically, the harmonization used for downstream evaluation reuses the exact training pipeline* (channel order, $[0, 1]$ normalization, slice-major 2.5D interleaving); a mismatched pipeline corrupts the generator input and can manufacture a spurious downstream regression.

Baselines / ablation. The λ_{NCE} sweep includes the cycle-only control ($\lambda_{\text{NCE}}=0$) on the same backbone, data, and budget, which is the primary ablation isolating the patch-contrastive term; a one-sided CUT-only baseline is left to future work.

5 Results

The contrastive weight has an interior optimum. Table 1 and Fig. 2 summarize the λ_{NCE} sweep. Three findings are consistent. First, Fréchet distance to the target site is *non-monotone* in λ_{NCE} with a clear minimum at 0.5 ($\text{FID}_{A \rightarrow B}=47.3$, versus 51.8 for the cycle-only control and 48.2 at $\lambda_{\text{NCE}}=2.0$): a moderate contrastive term improves distributional alignment, while over-weighting erodes it—the regime predicted if cycle-consistency supplies the global appearance anchor and PatchNCE acts as a complement rather than a replacement. Second, brain-masked structural preservation between source and harmonized image rises

Table 1. Harmonization metrics across the contrastive-weight sweep ($A \rightarrow B$, subject-level). Domain-classifier accuracy is reported raw $\rightarrow$ harmonized (chance = 0.5); FID and the source $\leftrightarrow$ harmonized masked windowed structural similarity are the primary axes. Best FID in bold.

λ_{NCE}	dom.-clf raw $\rightarrow$ harm	FID $_{A \rightarrow B}$	masked SSIM (cycle)	struct. SSIM	MMD
0	0.639 $\rightarrow$ 0.405	51.8	0.702	0.289	0.0007
0.1	0.639 $\rightarrow$ 0.484	51.5	0.703	0.322	0.0013
0.5	0.639 $\rightarrow$ 0.505	47.3	0.709	0.419	0.0011
1	0.639 $\rightarrow$ 0.503	48.5	0.714	0.420	0.0006
2	0.639 $\rightarrow$ 0.501	48.2	0.710	0.440	0.0007

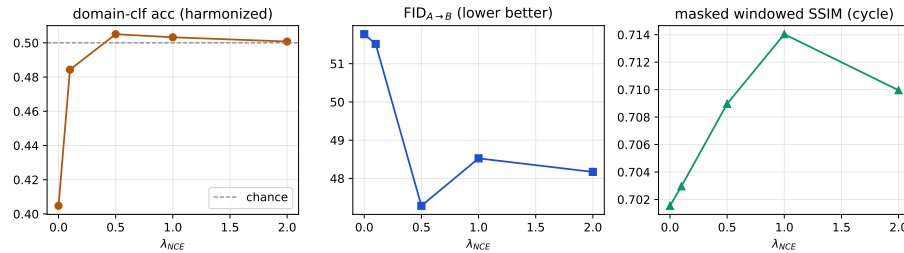


Fig. 2. Sweep trends. Left: domain-classifier accuracy reaches chance for $\lambda_{\text{NCE}} \geq 0.5$. Center: FID $_{A \rightarrow B}$ has an interior minimum at $\lambda_{\text{NCE}}=0.5$. Right: cycle-reconstruction SSIM remains high across the sweep.

monotonically with λ_{NCE} ($0.29 \rightarrow 0.44$), confirming that patch-contrastive alignment tightens the local content preservation that pure cycle-consistency leaves loose. Third, all hybrid arms drive the domain classifier to chance (≈ 0.50), whereas the cycle-only control *overshoots* below chance (0.41), consistent with an appearance-leaking equilibrium that the contrastive term regularizes. Cycle-reconstruction similarity stays high throughout (≈ 0.71), so these gains do not trade off against cycle consistency. The operating point $\lambda_{\text{NCE}}=0.5$ jointly optimizes distributional match and content preservation.

Downstream: a small cross-site gap that harmonization does not close. We next ask the decisive question—does harmonization help a downstream task? Table 2 and Fig. 3 report cross-site glioma segmentation with a U-Net trained on one site and applied to the other, raw versus harmonized. The first observation is that *the cross-site gap is already small*: a segmenter trained on site A scores 0.78 foreground Dice on raw site B , essentially matching its within-site performance (0.77 on A), and the reverse direction behaves similarly (0.77 cross-site versus 0.86 within-site). There is therefore little domain gap for image harmonization to recover. The second observation is that harmonization does *not* recover it and in fact lowers Dice in both directions (-53% for $A \rightarrow B$, -37% for $B \rightarrow A$ at $\lambda_{\text{NCE}}=0.5$). The effect is systematic rather than specific to one arm: the cycle-

Table 2. Cross-site segmentation, raw versus harmonized (foreground Dice, subject-level, $\lambda_{\text{NCE}}=0.5$). Within-site scores bound the achievable transfer. The already-small raw cross-site gap is not recovered by harmonization.

direction	raw Dice	harm. Dice	Δ	rel. %
BraTS $\rightarrow$ UPenn (train A , test B)	0.776	0.363	−0.414	−53.3
UPenn $\rightarrow$ BraTS (train B , test A)	0.765	0.479	−0.286	−37.4
within-site BraTS	0.765			
within-site UPenn	0.855			

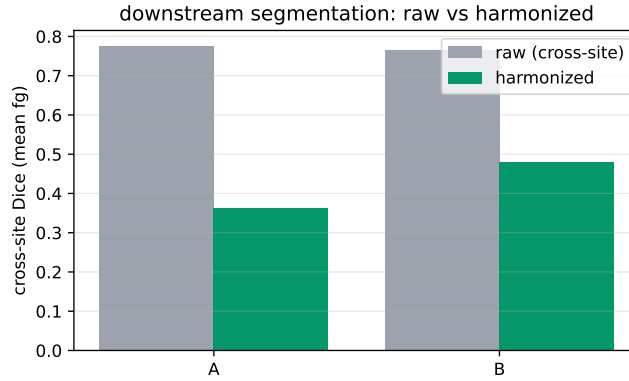


Fig. 3. Cross-site segmentation Dice, raw (gray) versus harmonized (green), in both transfer directions. The raw cross-site baseline is already close to the within-site ceiling, and harmonization reduces rather than improves it.

only control degrades by a comparable margin (−42% in both directions), so the loss is a property of image harmonization under this protocol, not of the contrastive term.

Why harmonization does not help here. Two factors explain the gap. (i) Both cohorts are glioma datasets with co-registered four-modality protocols normalized to $[0, 1]$, so the residual *segmentation-relevant* cross-site shift is small—image harmonization is expected to pay off precisely where it is least needed here. (ii) The segmenter is trained on real images and applied to GAN-synthesized images; a diagnostic comparison confirms harmonized volumes are anatomically faithful and retain tumor contrast, but they carry synthesis texture that is out-of-distribution for a real-image segmenter and redistribute per-modality contrast (e.g. weakening the FLAIR cue that delineates whole-tumor). This is a train/test shift introduced *by* harmonization, not a failure of anatomical preservation.

6 Discussion and Limitations

The mechanism question. The sweep supports the complementarity hypothesis: adding a moderate patch-contrastive term to cycle-consistency improves both distributional alignment (interior FID optimum at $\lambda_{\text{NCE}}=0.5$) and local structural preservation (monotone in λ_{NCE}), while over-weighting degrades distributional match—the signature of a complement that should not be allowed to dominate. The cycle-only control’s below-chance domain classifier is the appearance-leaking equilibrium made measurable, and the contrastive term regularizes it.

Distributional quality $\neq$ downstream utility. Our central negative result is methodological rather than a defect of the model. Harmonization that is excellent by FID, domain confusion, and masked structural similarity does not translate into downstream benefit when (a) the cross-site task gap is already small and (b) the downstream model is trained on real images and tested on synthesized ones. Crucially, we verified that the corrected, training-consistent harmonization pipeline (matched channel order, $[0, 1]$ normalization, slice-major 2.5D interleaving) produces anatomically faithful volumes—ruling out the mismatched-pipeline artifact that can manufacture a spurious regression—and the degradation nonetheless persists and is uniform across λ_{NCE} . The honest reading is that harmonization-quality metrics and task utility are distinct axes, and downstream evaluation must be reported even, and especially, when it disagrees with distribution metrics.

Limitations. (i) A single fine-tuning seed per arm; the FID ordering is consistent but per-arm variance is not estimated. (ii) Two-site, single-disease (glioma) scope: the small raw cross-site gap need not generalize to larger cross-scanner shifts where harmonization is known to help, and testing that regime is the natural next experiment. (iii) The downstream segmenter is trained only on real images; a train-on-harmonized or harmonization-as-augmentation protocol would disentangle the synthesis-texture penalty from any alignment benefit and is the primary journal-version extension. (iv) A one-sided CUT-only baseline is omitted; the cycle-only ($\lambda_{\text{NCE}}=0$) control isolates the contrastive term within our design but not against pure CUT.

7 Conclusion

We presented a mechanism study of cycle-consistent, patch-contrastive MRI harmonization on multi-site glioma data. A controlled contrastive-weight sweep shows that PatchNCE complements cycle-consistency—improving distributional alignment with an interior optimum at $\lambda_{\text{NCE}}=0.5$ and monotonically tightening structural preservation—without sacrificing cycle consistency. Under a corrected, training-consistent downstream protocol, however, the cross-site segmentation gap is already small and image harmonization does not close it, uniformly across λ_{NCE} . We conclude that distributional harmonization quality does not by itself

entail downstream task utility, and we argue for downstream-grounded, training-consistent evaluation as standard practice in image harmonization.

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